Incomplete Taxon Sampling via Data Augmentation in the Emphasis Importance Sampling Framework

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1 Problem Statement

The current `emphasis` importance sampling framework augments an observed phylogenetic tree T_{obs} with latent *extinct* lineages z_{ext} to compute the marginal likelihood:

$$L(\boldsymbol{\theta} \mid T_{\text{obs}}) = \int p(T_{\text{obs}}, z_{\text{ext}} \mid \boldsymbol{\theta})\, dz_{\text{ext}}. \tag{1}$$

This assumes that T_{obs} contains *all extant species* ($\rho = 1$). In practice, many phylogenies are incomplete: only a fraction $\rho \in (0, 1]$ of extant species are sampled. The N_{obs} tips in the observed tree represent only a subset of the true $N_{\text{tot}} = N_{\text{obs}}/\rho$ extant species. The remaining

$$m = N_{\text{tot}} - N_{\text{obs}} = N_{\text{obs}} \cdot \frac{1-\rho}{\rho} \tag{2}$$

species are extant but unsampled. These unsampled lineages are *not* extinct—they are alive at the present and contribute to diversity $N(t)$ throughout their existence. Ignoring them biases the likelihood, especially for diversity-dependent models where rates depend on $N(t)$.

2 Extended Augmentation Space

We extend the latent space to include both extinct lineages and unsampled extant lineages:

$$z = (z_{\text{ext}},\ z_{\text{unsamp}}) \tag{3}$$

where $z_{\text{ext}} = \{(s_j, e_j)\}_{j=1}^{n_e}$ are speciation–extinction time pairs with $e_j < T$, and $z_{\text{unsamp}} = \{s_k\}_{k=1}^{n_u}$ are speciation times of lineages that survive to the present but were not sampled. The marginal likelihood becomes:

$$L(\boldsymbol{\theta} \mid T_{\text{obs}}, \rho) = \int p(T_{\text{obs}}, z_{\text{ext}}, z_{\text{unsamp}} \mid \boldsymbol{\theta}, \rho)\, dz_{\text{ext}}\, dz_{\text{unsamp}}. \tag{4}$$

3 Complete-Data Likelihood

Let $\tilde{T} = (T_{\text{obs}}, z_{\text{ext}}, z_{\text{unsamp}})$ denote the fully augmented tree. This tree has:

- N_{obs} observed tips (extant, sampled),
- n_u unsampled tips (extant, not sampled),

- n_e extinct lineages (speciated and went extinct before T).

Let $N(t)$ denote the number of lineages alive at time t in $\tilde{T}$, which now includes the unsampled extant lineages.

The complete-data log-likelihood decomposes as:

$$\log p(\tilde{T} \mid \boldsymbol{\theta}, \rho) = \underbrace{\sum_{\text{spec}} \log \lambda_i + \sum_{\text{ext}} \log \mu_i - \int_0^T N(t)\big[\lambda(t) + \mu(t)\big]\,dt}_{\text{birth–death process}} + \underbrace{N_{\text{obs}} \log \rho + n_u \log(1-\rho)}_{\text{sampling at present}}. \quad (5)$$

The first part is the standard birth–death likelihood over the complete tree. The sampling terms arise because each extant lineage at time T is independently sampled with probability ρ or missed with probability $1-\rho$.

Key point. The unsampled species affect $N(t)$ from their speciation time s_k to the present T. For diversity-dependent models ($\lambda = \lambda(N)$), this changes the speciation and extinction rates throughout the tree. This is why simply adjusting the thinning rate analytically (the current approach) is insufficient—the unsampled lineages must be *explicitly placed* in the tree so that $N(t)$ is correct.

4 Proposal Distribution

4.1 Current Approach (Recap)

The thinning algorithm sweeps from root to present. In each interval $[t_k, t_{k+1})$ between consecutive events:

1. Compute the thinning envelope $\Lambda_{\max} = \max_{t \in [t_k, t_{k+1}]} N(t) \cdot \lambda(t; \boldsymbol{\theta}) \cdot (1 - \rho\, e^{-\mu(t)(T-t)})$.
2. Propose next event: $t^* = t_k - \log(u_1)/\Lambda_{\max}$, $u_1 \sim \text{Unif}(0,1)$.
3. If $t^* < t_{k+1}$, accept with probability $p_{\text{thin}} = N(t^*)\lambda(t^*)(1 - \rho\, e^{-\mu(t^*)(T-t^*)})/\Lambda_{\max}$.
4. If accepted, draw extinction time $e^* \sim \text{TruncExp}(\mu(t^*), T - t^*)$.

Currently, if $e^* \geq T$ (lineage survives), the event is *skipped* and not inserted into the tree.

4.2 New Approach: Insert Unsampled Extant Species

We modify step 4. After accepting a speciation event at t^*:

4a. Draw whether the new lineage goes extinct or survives. The probability that a lineage born at t^* goes extinct before T is:

$$P_{\text{ext}}(t^*) = 1 - e^{-\mu(t^*)(T-t^*)}. \quad (6)$$

With incomplete sampling, a surviving lineage is unsampled with probability $1-\rho$. So the three outcomes and their probabilities are:

$$\text{Extinct:} \quad P_{\text{ext}}(t^*) = 1 - e^{-\mu(T-t^*)} \quad (7)$$

$$\text{Extant, unsampled:} \quad (1-\rho) \cdot e^{-\mu(T-t^*)} \quad (8)$$

$$\text{Extant, sampled:} \quad \rho \cdot e^{-\mu(T-t^*)} \quad \text{(impossible—already observed)} \quad (9)$$

The thinning rate $N\lambda(1-\rho\, e^{-\mu(T-t)})$ already excludes the "extant, sampled" outcome, so conditionally:

$$P(\text{extinct} \mid \text{accepted}) = \frac{1-e^{-\mu(T-t^*)}}{1-\rho\, e^{-\mu(T-t^*)}} \tag{10}$$

$$P(\text{unsampled extant} \mid \text{accepted}) = \frac{(1-\rho)\, e^{-\mu(T-t^*)}}{1-\rho\, e^{-\mu(T-t^*)}}. \tag{11}$$

4b. **If extinct:** draw $e^* \sim \text{TruncExp}(\mu, T-t^*)$ and insert both a speciation node at t^* and an extinction node at e^*. Update $N(t)$ for $t \in [t^*, e^*)$.

4c. **If unsampled extant:** insert a speciation node at t^* with the lineage surviving to T. Update $N(t)$ for $t \in [t^*, T]$ (increment by 1 for all subsequent events).

Remark. This is precisely what the existing `extinction_time()` function already computes: it returns $e^* \geq T$ with probability $(1-\rho)e^{-\mu(T-t^*)}/(1-\rho\, e^{-\mu(T-t^*)})$. The only change is that when $e^* \geq T$, we *insert* the unsampled species into the tree (via `insert_unsampled_species()`) instead of skipping it.

5 Sampling Probability $\log q(z \mid T_{\text{obs}}, \boldsymbol{\theta}, \rho)$

The proposal probability q for the augmented tree decomposes into contributions from the thinning process (the integral term) and from each inserted lineage:

$$\log q = \sum_{j\in\mathcal{E}} \log q_j^{\text{ext}} + \sum_{k\in\mathcal{U}} \log q_k^{\text{unsamp}} - \int_0^T N(t)\lambda(t)\big(1-\rho\, e^{-\mu(t)(T-t)}\big)\, dt \tag{12}$$

where $\mathcal{E}$ indexes extinct augmented lineages and $\mathcal{U}$ indexes unsampled extant augmented lineages.

For an **extinct** augmented lineage j with speciation at s_j, extinction at e_j, parent chosen uniformly from K_j alive lineages:

$$\log q_j^{\text{ext}} = \log\big(N(s_j)\cdot\lambda(s_j)\cdot\mu(s_j)\big) - \mu(s_j)(e_j - s_j) - \log K_j + \log \frac{1-e^{-\mu(s_j)(T-s_j)}}{1-\rho\, e^{-\mu(s_j)(T-s_j)}}. \tag{13}$$

The last term is the log-probability of choosing the "extinct" branch (step 4a), conditional on acceptance.

For an **unsampled extant** lineage k with speciation at s_k:

$$\log q_k^{\text{unsamp}} = \log\big(N(s_k)\cdot\lambda(s_k)\big) - \log K_k + \log \frac{(1-\rho)\, e^{-\mu(s_k)(T-s_k)}}{1-\rho\, e^{-\mu(s_k)(T-s_k)}}. \tag{14}$$

Simplification when $\rho = 1$. When $\rho = 1$, the unsampled probability is zero, the conditional extinct probability is 1, and Eq. (**??**) reduces to the current implementation. The thinning rate becomes $N\lambda(1-e^{-\mu(T-t)})$ and the per-lineage terms simplify to $\log(N\lambda\mu) - \mu\ell - \log K$, recovering the existing `sampling_prob()`.

6 Importance Weight

The importance weight for augmented tree i is:

$$w_i = \exp\bigl(\log f_i - \log g_i\bigr) \tag{15}$$

where $\log f_i = \log p(\tilde{T}_i \mid \boldsymbol{\theta}, \rho)$ from Eq. (**??**) and $\log g_i = \log q(z_i \mid T_{\text{obs}}, \boldsymbol{\theta}, \rho)$ from Eq. (**??**). The log-likelihood estimate is:

$$\hat{\ell}(\boldsymbol{\theta}) = \log \frac{1}{S} \sum_{i=1}^{S} w_i. \tag{16}$$

7 Implementation Strategy

The required changes are minimal because the infrastructure already exists:

1. `augment_tree.cpp`, `do_augment_tree()`: When `extinction_time()` returns $e^* \geq T$, call `insert_unsampled_species()` instead of skipping. This function already exists and correctly updates $N(t)$ for all subsequent nodes.

2. `model.hpp`, `loglik()`: Add the sampling terms $N_{\text{obs}} \log \rho + n_u \log(1-\rho)$ to the log-likelihood. Also ensure that unsampled extant lineages (identified by `is_unsampled(node)`) contribute to the integral via their effect on $N(t)$ but do *not* generate an extinction event term.

3. `model.hpp`, `sampling_prob()`: Update the per-lineage contribution for unsampled extant lineages to match Eq. (**??**). The existing `is_unsampled()` branch needs correction: include the conditional probability term $\log \frac{(1-\rho)e^{-\mu(T-s)}}{1-\rho\, e^{-\mu(T-s)}}$ and remove the bare $\mu(T-s)$ subtraction. Similarly update the extinct branch to include $\log \frac{1-e^{-\mu(T-s)}}{1-\rho\, e^{-\mu(T-s)}}$.

4. **R layer**: No changes needed—`rho` is already threaded through `simulate_tree()`, `estimate_rates()`, and all control lists.

Default behavior. When $\rho = 1$ (default), $m = 0$, the unsampled probability is zero, `extinction_time()` never returns $e^* \geq T$, and the algorithm reduces exactly to the current implementation.

8 Algorithm Summary

Algorithm 1 Tree augmentation with incomplete taxon sampling

Require: Observed tree T_{obs}, parameters $\boldsymbol{\theta}$, sampling fraction ρ

Initialize augmented tree $\tilde{T} \leftarrow T_{\text{obs}}$
$t \leftarrow 0$
while $t < T$ **do**
  $t_{\text{next}} \leftarrow$ next event time in $\tilde{T}$ after t
  $\Lambda_{\max} \leftarrow \max_{s \in [t, t_{\text{next}}]} N(s)\lambda(s; \boldsymbol{\theta})(1 - \rho\, e^{-\mu(s)(T-s)})$
  $t^* \leftarrow t - \log(u_1)/\Lambda_{\max}$
  if $t^* < t_{\text{next}}$ **then**
    $p_{\text{thin}} \leftarrow N(t^*)\lambda(t^*)(1 - \rho\, e^{-\mu(t^*)(T-t^*)})/\Lambda_{\max}$
    if $u_2 < p_{\text{thin}}$ **then** ▷ Accept speciation
      Draw e^* from $\textsc{ExtinctionTime}(t^*, \mu, T, \rho)$
      if $e^* < T$ **then** ▷ Extinct lineage
        Insert speciation at t^*, extinction at e^* into $\tilde{T}$
        Update $N(t)$ for $t \in [t^*, e^*)$
      else ▷ Unsampled extant lineage
        Insert speciation at t^* into $\tilde{T}$ (survives to T)
        Update $N(t)$ for $t \in [t^*, T]$
      end if
    end if
  end if
  $t \leftarrow \min(t^*, t_{\text{next}})$
end while
Compute pendant PD for all nodes
return $\tilde{T}$